# Math 105L In Class 2 Practice

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# Limits and Continuity

1. Describe the three ways in which a function can fail to be continuous. Draw an example in each case.

2. We said that $f(x)$ is *continuous* at $x = a$ if $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = f(a)$.

   We also said that $f(x)$ is continuous at $x = a$ if we both *know what to expect at* $x = a$ and *get what we expect at* $x=a$.

   Of the two equalities in the first definition, which says we know what to expect, and which says we get it? Explain.

3. Consider the function $f(x) = \frac{x^2-1}{x+1}$.

   (a) Explain why $f(x)$ is not continuous at $x = -1$.

   (b) Consider

   $$g(x) = \begin{cases} f(x) & x \neq -1 \\ k & x = -1 \end{cases}$$

   Is there a value of $k$ that makes $g(x)$ continuous? If so, find it. If not, explain why.

4. Consider the function

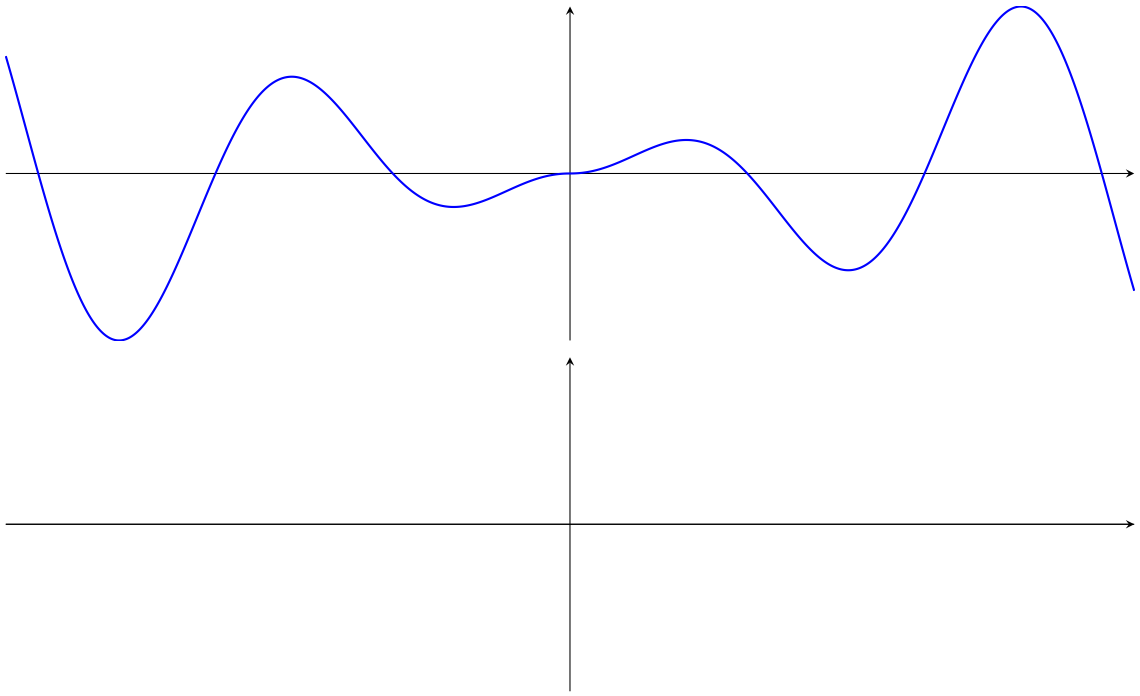
   $$f(x) = \begin{cases} -1 & x > 0 \\ k & x = 0 \\ 1 & x < 0 \end{cases}$$

   Is there a value of $k$ that makes $f(x)$ continuous? If so, find it. If not, explain why.

5. Briefly explain why it makes sense that $\lim_{x\to a} kf(x) = k \lim_{x\to a} f(x)$.

# Derivative Function

1. Sketch the graph of the derivative of the following function on the axes below.



2. Answer each of the following questions regarding the relationship between the sign/behavior of a function and its derivative:

   (a) If $f(x)$ is positive for $0 < x < 3$, and negative for $3 < x < 6$, what is the sign of its derivative for $3 < x < 6$? Explain.

   (b) If $g'(x) > 0$ for all $1 < x < 5$, does this guarantee that $g(x) > 0$ for all $1 < x < 5$? Explain.

   (c) Suppose $h(x)$ is continuous and differentiable for $0 < x < 8$. If $h'(x) < 0$ for $0 < x < 8$ and $h(3) = 0$, what can you say about the sign of $h(x)$ for $x < 3$? What about for $x > 3$? Explain.

3. Explain the practical meaning of each of the following statements, including units:

   (a) Suppose that $F(p)$ is a function giving the number of thousands of fish in a county with $p$ ponds in it. Explain what the following quantities mean:

   (i) $F(20) = 100$.
   (ii) $F'(20) = 6$.

   (b) Let $C(w)$ be the monthly cost (in dollars) of heating a home when the average outside temperature during that month is $w$ degrees Fahrenheit (°F). Explain what the following quantities mean:

   (i) $C(32) = 210$.
   (ii) $C'(32) = -6$.

(c) Let $M(t)$ be the total mass of honey (in kilograms) produced by a bee colony $t$ weeks after the start of spring. Explain what the following quantities mean:

(i) $M(8) = 15$.

(ii) $M'(8) = 2.4$.

4. Answer each of the following questions regarding physical interpretations of functions and rates of change:

(a) Let $A(t)$ be the depth below sea level of a submarine $t$ hours after its launch, in feet.

(i) Explain why the statement $A(5) = -1000$ makes no physical sense. Include units in your answer.

(ii) Suppose that $A'(5) = -500$. Discuss the motion of the submarine at this time, including direction, speed, and units.

(b) Let $H(t)$ be the altitude above ground level of a weather balloon, in meters, $t$ minutes after release.

(i) Explain why the statement $H(12) = -80$ makes no physical sense. Include units.

(ii) Suppose that $H'(12) = -5$. Discuss the motion of the balloon at this time, including direction, speed, and units.

(c) Let $V(t)$ be the volume of water (in gallons) remaining in an agricultural storage reservoir $t$ days after the start of a drought. (Assume the reservoir cannot have a negative volume.)

(i) Explain why the statement $V(40) = -500$ makes no practical sense. Include units.

(ii) Suppose that $V'(40) = -120$. Discuss what is happening to the reservoir at this time, including rate and units.

5. For each of the following functions, compute the derivative $f'(x)$ *from the definition of the derivative, without using any differentiation rules* ($f'(x) = \lim_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$):

(a) $f(x) = x^2 + 2x$

(b) $f(x) = 3x^2 - 5x + 1$

(c) $f(x) = \frac{1}{x+2}$

# Second Derivative

1. If $f''(x) > 0$, then which of the following is true? Explain why each is true, false, or unknowable.

   (a) The function $f(x)$ is positive.

   (b) The tangent lines to $f(x)$ are increasing.

   (c) The slopes of the tangent lines to $f(x)$ are increasing.

2. Sketch graphs of smooth continuous functions with domain all real numbers matching each one of the following descriptions, or explain why it's impossible.

   (a) $f(x) > 0$, $f'(x) < 0$, $f''(x) < 0$, and $\lim_{x\to\infty} = 5$.

   (b) $f(x)$ has two inflection points, $f'(x) < 0$ for $x < 0$, $f'(x) \geq 0$ for $x > 0$.

   (c) $f(x)$ has no inflection points, $f'(x) = 0$ at exactly one point, $f(x)$ has two zeros.

3. True or False? Explain your answers.

   (a) If a function is concave up, then $f''(x) > 0$.

   (b) If a function is concave down, then $f''(x) \leq 0$.

   (c) When a function turns around, it has an inflection point.

   (d) It is possible for a function to be decreasing, but have increasing slopes.

4. Fill in the blanks: If a function is increasing, then its _____ is positive. If the slopes of a function are decreasing, then its _____ is negative.

5. The position of an object above the ground at time $t$ is given by a function $s(t)$, where $s'(t) < 0$ and $s''(t) > 0$. Is the object rising or falling? Is it speeding up or slowing down? Explain.

# Product and Quotient Rules

1. Differentiate $f(x) = x^2(3x^3 + 1)$.
2. Differentiate $g(x) = (x^3 - 2x)(x + 5)$.
3. Differentiate $h(x) = \sqrt{x}\,(2x^2 - 1)$. (Assume $x > 0$.)
4. Differentiate $p(x) = (x - 1)^2(x^2 + 2)$.
5. Differentiate $q(x) = e^x(x^2 - 3x + 1)$.
6. Differentiate $r(x) = \dfrac{x^2 + 1}{x}$.
7. Differentiate $s(x) = \dfrac{3x - 1}{x^2 + 1}$.
8. Differentiate $t(x) = \dfrac{x^3 + 2}{x - 1}$.
9. Differentiate $u(x) = \dfrac{\sqrt{x}}{x + 2}$.
10. Differentiate $v(x) = (x^2 + 1)(2x^3 - x)$.
11. Find the tangent line to $f(x) = x^2(3x + 1)$ at $x = 1$.
12. Find the tangent line to $g(x) = (x^3 - 2x)(x + 2)$ at $x = 0$.
13. Find the tangent line to $h(x) = \sqrt{x}(2x^2 - 1)$ at $x = 1$.
14. Find the tangent line to $p(x) = (x - 1)^2(x^2 + 1)$ at $x = 2$.
15. Find the tangent line to $q(x) = e^x(x^2 - 2x)$ at $x = 0$.
16. Find the tangent line to $r(x) = \frac{x^2+1}{x}$ at $x = 1$.
17. Find the tangent line to $s(x) = \frac{3x-1}{x^2+1}$ at $x = 1$.
18. Find the tangent line to $t(x) = \frac{x^3+1}{x-1}$ at $x = 2$.
19. Find the tangent line to $u(x) = \frac{\sqrt{x}}{x+1}$ at $x = 4$.
20. Find the tangent line to $v(x) = (x^2 + 1)(2x^2 - x)$ at $x = -1$.

# Inverse Functions and Composition

1. Explain the meaning of each of the following statements in practical terms:

   (a) Let $P(x)$ be the number of elephants within a distance of $x$ meters from my bathtub. Explain the meaning of each of the following in practical terms, including units:

   (i) $P(20) = 4$
   (ii) $P^{-1}(10) = 55$
   (iii) $P^{-1}(0) = 8$

   (b) A factory produces custom bicycles. Let $C(n)$ be the total cost (in dollars) of producing $n$ bicycles in a day, and let $N(t)$ be the number of bicycles produced after the factory has been running for $t$ hours. Explain the meaning of each of the following in practical terms:

   (i) $C(50) = 15{,}000$
   (ii) $C^{-1}(8000) = 25$
   (iii) The composite function $K(t) = C(N(t))$
   (iv) $K^{-1}(12{,}000) = 4$
   (v) Explain why the expression $N(C(50))$ does not make physical sense.

   (c) Let $T(h)$ be the temperature (in °C) of the atmosphere at an altitude of $h$ kilometers above sea level, and let $D(T)$ be the density of air (in kg/m$^3$) when the air temperature is $T$°C. Explain the meaning of each of the following, including units:

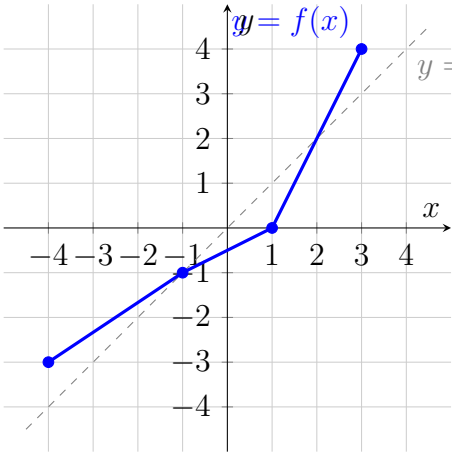
   (i) $T(4) = -11$
   (ii) $T^{-1}(0) = 2.5$
   (iii) The composite function $\rho(h) = D(T(h))$
   (iv) $\rho^{-1}(0.9) = 3$

   (d) A patient is administered a medication. Let $C(t)$ be the concentration of the drug in the bloodstream (in mg/L) $t$ hours after administration, and let $R(c)$ be the patient's pain relief score (on a scale from 0 to 100) at drug concentration $c$ mg/L. Explain the meaning of each of the following in practical terms:

   (i) $C(2) = 18$
   (ii) $C^{-1}(5) = 9$
   (iii) The composite function $S(t) = R(C(t))$
   (iv) $S^{-1}(80) = 3$

2. For each of the following functions $y = f(x)$ shown on the axes below, sketch the graph of the inverse function $y = f^{-1}(x)$ and answer the associated questions:

   (a) The graph of $y = f(x)$ is shown below, with domain $-4 \leq x \leq 3$ and range $-3 \leq y \leq 4$.

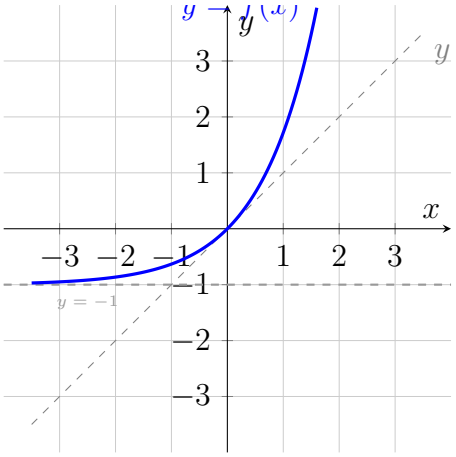



   (i) On the axes above, sketch the graph of the inverse function $y = f^{-1}(x)$.

   (ii) Given that the domain of $f(x)$ is $-4 \leq x \leq 3$ and its range is $-3 \leq y \leq 4$, state the domain and range of $f^{-1}(x)$.

   (iii) At what point do the graphs of $f(x)$ and $f^{-1}(x)$ intersect?

   (b) The graph of $f(x) = e^x - 1$ is shown below, with domain $-\infty < x < \infty$ and range $y > -1$. The horizontal asymptote at $y = -1$ is indicated by a dashed line.

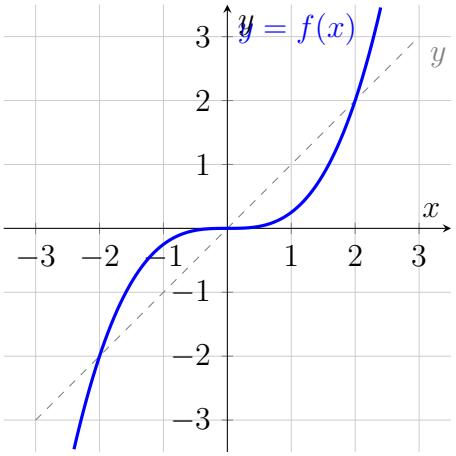



   (i) What asymptote does the inverse function $y = f^{-1}(x)$ have? Write its equation.

   (ii) On the axes above, sketch the graph of $y = f^{-1}(x)$.

   (iii) Given that the domain of $f(x)$ is $-\infty < x < \infty$ and its range is $y > -1$, state the domain and range of $f^{-1}(x)$.

(c) The graph of $f(x) = \frac{1}{4}x^3$ is shown below along with the diagonal line $y = x$, with domain $-\infty < x < \infty$ and range $-\infty < y < \infty$.

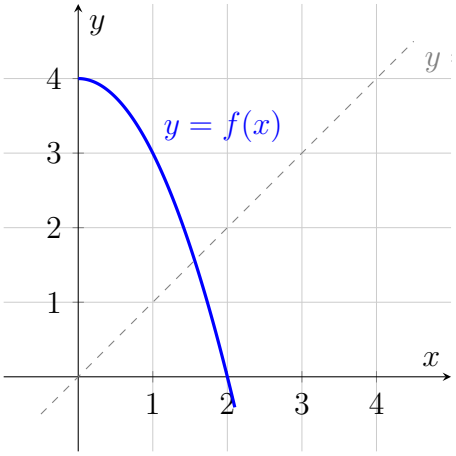



(i) On the axes above, sketch the graph of the inverse function $y = f^{-1}(x)$.

(ii) Given that the domain of $f(x)$ is $-\infty < x < \infty$ and its range is $-\infty < y < \infty$, state the domain and range of $f^{-1}(x)$.

(iii) The graph of $f(x)$ has a horizontal tangent line at the origin $(0, 0)$. What is the tangent line to the graph of $f^{-1}(x)$ at the origin? Explain geometrically.

(d) The graph below shows the function $f(x) = 4 - x^2$ defined on the restricted domain $x \geq 0$, where its range is $y \leq 4$.




(i) Explain why $f(x) = 4 - x^2$ on $-\infty < x < \infty$ is not invertible, but becomes invertible when its domain is restricted to $x \geq 0$.

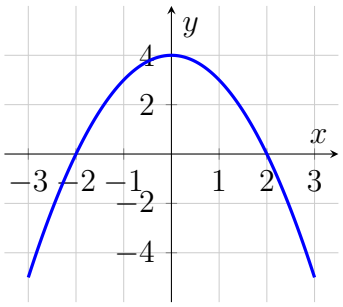
(ii) On the axes above, sketch the graph of $y = f^{-1}(x)$.

(iii) Given that the domain of this restricted function $f(x)$ is $x \geq 0$ and its range is $y \leq 4$, state the domain and range of $f^{-1}(x)$.

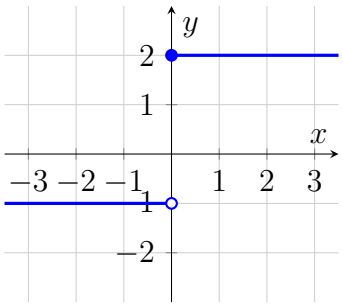
3. For each pair of functions $f$ and $g$ shown below, sketch the graphs of the composite functions $y = f(g(x))$ and $y = g(f(x))$:

(a) The graphs of $f(x) = 4 - x^2$ and $g(x) = \begin{cases} -1 & x < 0 \\ 2 & x \geq 0 \end{cases}$ are given below.

$y = f(x)$




$y = g(x)$

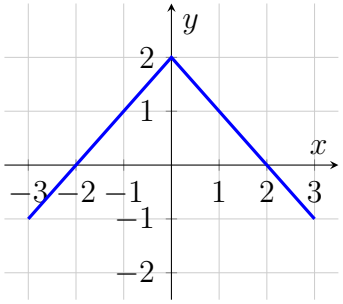



(i) Sketch the graph of $y = f(g(x))$. Explain your reasoning.

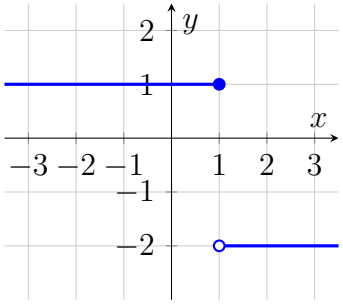
(ii) Sketch the graph of $y = g(f(x))$. Explain your reasoning.

(b) The graphs of $f(x) = 2 - |x|$ and $g(x) = \begin{cases} 1 & x \leq 1 \\ -2 & x > 1 \end{cases}$ are shown below.

$y = f(x)$



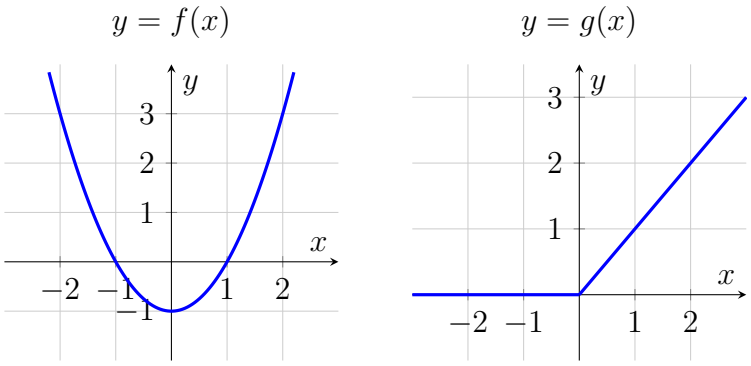

$y = g(x)$




(i) Sketch the graph of $y = f(g(x))$. Explain your reasoning.

(ii) Sketch the graph of $y = g(f(x))$. Explain your reasoning.

(c) **[Challenge]** The graphs of $f(x) = x^2 - 1$ and the ramp function $g(x) = \begin{cases} 0 & x < 0 \\ x & x \geq 0 \end{cases}$ are given below.




(i) Write a piecewise formula for $f(g(x))$, and sketch its graph.

(ii) Write a piecewise formula for $g(f(x))$, and sketch its graph.

4. For each of the following functions, find a formula for the inverse function and state its domain and range:

(a) Consider the rational function $f(x) = \frac{3x+2}{x-4}$, which has domain $x \neq 4$ and range $y \neq 3$.

(i) Find a formula for $f^{-1}(x)$. Show all your algebraic steps clearly.

(ii) State the domain and range of $f^{-1}(x)$.

(b) Consider the exponential function $g(x) = 5 - 2e^{3x-1}$, which has domain $-\infty < x < \infty$ and range $y < 5$.

(i) Find a formula for $g^{-1}(x)$. Show all your steps.

(ii) State the domain and range of $g^{-1}(x)$.

(c) Consider the logarithmic function $h(x) = 4\ln(2x-6)+1$, which has domain $x > 3$ and range $-\infty < y < \infty$.

(i) Find a formula for $h^{-1}(x)$, showing all your work.

(ii) State the domain and range of $h^{-1}(x)$.

(d) Consider the base-10 exponential function $k(x) = 2 \cdot 10^{3x} - 4$, which has domain $-\infty < x < \infty$ and range $y > -4$.

(i) Find a formula for $k^{-1}(x)$. Show all your steps.

(ii) State the domain and range of $k^{-1}(x)$.

# Compound Interest

1. Which of the following is good approximation of $Ae^{0.04t}$? Explain your answer.

   (a) $A\left(1+\frac{1}{0.04}\right)^{1,000,000}$

   (b) $A\left(1+\frac{0.04}{1,000,000}\right)^{1,000,000}$

   (c) $A\left(1+\frac{1,000,0000}{0.04}\right)^{1,000,000}$

   (d) $A\left(1+\frac{1}{0.04}\right)^{0.04}$

2. "The more often you compound, the more money you make, so it's worth insisting on an initial investment of \$1,000 being compounded every second rather than every minute."

   Critique this statement.

3. How much money would you have if you invested \$1000 at 50% annual interest, compounded twice a year? Don't use a formula here: show computations from scratch.

4. Write an equation whose solution answers each of the following continuous compounding questions:

   (a) Write an equation whose solution gives the initial amount of money $P_0$ you would need to invest to have \$100,000 in 20 years, assuming an account paying continuously compounded 6% annual interest.

   (b) Write an equation whose solution gives the annual interest rate $r$ (compounded continuously) needed for an initial deposit of \$4,000 to grow to \$10,000 in 15 years.

   (c) Write an equation whose solution gives the number of years $t$ it will take for an initial investment of \$2,500 to triple in value in an account paying 4.8% interest compounded continuously.

5. Suppose that you have the opportunity either get \$10,000 in ten years, or \$5,000 now, given that you can put the money in a bank account with 8% interest, compounded continuously. Which would you prefer? Explain.

# Exponential Functions

1. Which of the following functions is exponential?

   (a) $f(x) = 3x + 2$
   (b) $f(x) = x^3$
   (c) $f(x) = 2^x$
   (d) $f(x) = e^{x-1}$
   (e) $f(x) = 7^{x^2}$
   (f) $f(x) = |x| + 7$
   (g) $f(x) = x^x$
   (h) $f(x) = e^x + x$

2. Suppose that $f(x) = P \cdot a^x$, with $a > 0$ and $a \neq 1$. Which of the following are true regardless of the values of $P$ and $a$ (subject to $a > 0$, $a \neq 1$)? For those that are not always true, give values of $P$ and $a$ that make them true.

   (a) $f(x)$ is increasing.
   (b) $f(x)$ is decreasing.
   (c) $f(x)$ is concave up.
   (d) $f(x)$ is concave down.

3. Write each of the following functions in the form $P_0a^t$. Identify the initial value $P_0$ and the growth/decay factor $a$ in exact form, and provide a decimal approximation for $a$ to 3 decimal places:

   (a) $f(t) = 22e^{0.2t}$
   (b) $g(t) = 85e^{-0.05t}$
   (c) $h(t) = 4.5e^{1.2t}$

4. For each of the following scenarios, find an exponential formula of the form $P(t) = P_0a^t$ from the given data and answer the question. Show all your work clearly without using shortcut formulas:

   (a) Suppose that a radioactive substance has a half-life of 4 days. If we start with 20 moles, find a formula of the form $P_0a^t$ for the number of moles left at time $t$ days, and use it to find how much is left after 6 days.
   (b) A culture of bacteria doubles in population every 5 hours. If the culture initially contains 300 bacteria, find a formula of the form $P(t) = P_0a^t$ for the number of bacteria after $t$ hours, and use it to determine the population after 14 hours.
   (c) A prescribed medication has a half-life of 8 hours in the bloodstream. If a patient is administered an initial dose of 80 mg, find a formula of the form $Q(t) = P_0a^t$ for the amount of medication (in mg) remaining in the body after $t$ hours, and use it to find the amount remaining after 20 hours.

5. Suppose that $a > 0$. What coordinate do the graphs of all functions of the form $f(x) = a^x$ pass through?

# Logs

1. Find the quadrupling time of a bank account with 6% continuously compounded interest.

2. (a) Find the inverse of $f(x) = 3e^{2x} - 1$.
   (b) Find the inverse of $g(x) = \ln(5x - 2)$.
   (c) Find the inverse of $h(x) = 2^{x+3} + 4$.
   (d) Find the inverse of $p(x) = \frac{1}{4}\ln(x) + 7$.
   (e) Find the inverse of $q(x) = \ln\big(2 - e^{-x}\big)$.

3. (a) Solve $5^x = 12$.
   (b) Solve $10^{2x+1} = 7 \cdot 10^{x-2}$.
   (c) Solve $9^{7-2x} + 6 = e$.
   (d) Solve $\log(2x + 3) + \log(x - 1) = 1$. (Give exact answer and check if your answers are possible!)
   (e) Solve $\log(4x) - \log(x + 1) = \frac{1}{2}$. (Give exact answer and check if your answers are possible.)

4. Let $x = \log A$ and $y = \log B$. Rewrite each of the following in terms of $x$ and $y$:
   (a) $\log\sqrt{\dfrac{A}{B}}$
   (b) $\log\left(\dfrac{A^3\sqrt{B}}{10}\right)$
   (c) $\log\left(\dfrac{\sqrt[3]{A}\,B^{-2}}{A^{-1}}\right)$
   (d) $\dfrac{\log\big(A^2B\big)}{\log\big(\sqrt{AB}\,\big)}$

5. Write $9^x$ in the form $8^{kx}$.

# Chain Rule

1. Find the derivatives of the following functions:

   (a) $y = (2x^3 - x)^5$.
   (b) $y = e^{3x^2+5}$.
   (c) $y = ((x^2+1)^3 + 2)^4$.
   (d) $y = (e^{x^3+3x^2+2x})^2$.
   (e) $y = (x^4 + 2x^2 + 3)^{1/3}$.
   (f) $y = e^{(4x^3+1)^{1/2}}$.
   (g) $y = (5x^4 - 7x^2 + 2)^3$.
   (h) $y = ((x^2+1)^4 + 3x)^5$.
   (i) $y = e^{(x^2+1)^3}$.
   (j) $y = (3x^2 + (x+1)^3)^4$.

2. Supose that $f(x)$ is a differentiable function. Differentiate the following. Most involve the triple chain rule.

   (a) $f(x)^9$
   (b) $f\big((x^4+2)^5\big)$
   (c) $e^{f(x)^2+1}$
   (d) $2^{f(x)^3}$
   (e) $\big(f(x)^4 + x\big)^8$
   (f) $f\big((x^2+1)^7\big)$
   (g) $(10^{f(x)} + 1)^5$
   (h) $f\big(e^{x^3+2x}\big)$
   (i) $(f(x)^5 + 2)^9$
   (j) $f\big((x^4+3x^2+1)^3\big)$

3. Let $f(x)$ and $g(x)$ be differentiable. Find a formula for the derivative of $\frac{f(x)}{(g(x))^2}$. Then ***use it*** to compute the derivative of $\frac{e^x}{x^4}$.

4. For each of the following real-world composite scenarios, interpret the functions and apply the Chain Rule:

   (a) Suppose that the depth of a lake (in feet) at a distance of $d$ feet from its center is given by a function $f(d)$. Suppose also that the density of fish (in fish/ft$^3$) in an area of the lake with depth $D$ feet is given by a function $p(D)$.

      (i) Explain what the composite function $F(d) = p(f(d))$ means in practical terms.
      (ii) Explain why the expression $f(p(d))$ does not make physical sense.
      (iii) What are the units of $f'(d)$, $p'(D)$, and $F'(d)$?
      (iv) Suppose that at a distance of 1,000 feet from the center of the lake, the depth is 20 feet and is getting shallower at a rate of 0.1 ft of depth for each additional foot from the center. Furthermore, when the depth is 20 ft, each additional foot of depth increases the density of fish by 100 fish/ft$^3$. Compute the rate at which the density of fish changes per foot of distance from the center when you are 1,000 feet from the center. Give units.

(b) A research balloon ascends into the atmosphere. Let $h(t)$ be the altitude of the balloon (in kilometers) $t$ hours after launch, and let $T(h)$ be the atmospheric temperature (in °C) at altitude $h$ kilometers above sea level.

(i) Explain what the composite function $C(t) = T(h(t))$ means in practical terms.

(ii) Explain why the expression $h(T(t))$ does not make physical sense.

(iii) State the units of $h'(t)$, $T'(h)$, and $C'(t)$.

(iv) Suppose that 2 hours after launch, the balloon is at an altitude of 6 km and is climbing at a rate of 1.5 km/hr. At an altitude of 6 km, atmospheric temperature decreases at a rate of 6.5°C per kilometer of altitude increase. Compute $C'(2)$, the rate of change of the temperature experienced by the balloon with respect to time at $t = 2$ hours. Include units.

(c) An electric car's performance is tested. Let $s(t)$ be the speed of the car (in miles per hour) $t$ minutes after entering a highway, and let $E(s)$ be the vehicle's energy consumption rate (in watt-hours per mile, Wh/mi) when traveling at speed $s$ mph.

(i) Explain what the composite function $P(t) = E(s(t))$ represents in practical terms.

(ii) Explain why the expression $s(E(t))$ does not make physical sense.

(iii) State the units of $s'(t)$, $E'(s)$, and $P'(t)$.

(iv) Suppose that 10 minutes into the drive, the car is travelling at 65 mph and accelerating at a rate of 1.2 mph per minute. At a speed of 65 mph, aerodynamic drag causes the energy consumption rate to increase by 4 Wh/mi for every 1 mph increase in speed. Compute $P'(10)$ and include units.

# Differentiating Exponentials

Note: some of the chain rule problems above cover derivatives of exponentials.

1. Fill in the blanks: Let $a > 0$. Then the derivative of $f(x) = a^x$ is _____ proportional to $f(x)$ with constant of proportionality _____.

2. Suppose that $e^{g(x)} = x$. By differentiating both side of this equation, show that $g'(x) = \frac{1}{x}$.

3. Find the slope of the tangent line to each of the following functions at the given value of $x$:

   (a) $f(x) = 2^x + e^x + e^2 + 2^e + x^2 + x^e$ at $x = 1$.
   (b) $g(x) = 3^x - 2e^x + x^3 + 3^e$ at $x = 0$.
   (c) $h(x) = 5 \cdot 2^x + 4e^x - 3x^2 + \pi^2$ at $x = 1$.
   (d) $p(x) = 10^x - x^{10} + 10^{10} + e^x$ at $x = 1$.
   (e) $q(x) = 4e^x + 5^x + x^5 + 5^5$ at $x = 0$.
   (f) $r(x) = 2 \cdot 3^x + e^x - e \cdot x^{e-1} + e^\pi$ at $x = 1$.

4. Find a function of the form $f(x) = a^x$, so that $f'(x) = -f(x)$.

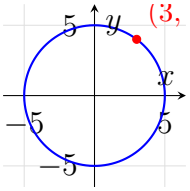
5. What is the exact value of $\lim_{h \to 0} \dfrac{7^h - 1}{h}$? Explain.

6. Find the derivative of each of the following exponential functions. Simplify your answers where appropriate:

   (a) $f(x) = 5e^x - 4 \cdot 3^x$
   (b) $g(x) = e^{5x-2}$
   (c) $h(x) = 7^{3x}$
   (d) $k(x) = x^4 e^x$
   (e) $m(x) = \frac{e^x}{x^2+3}$
   (f) $n(x) = 3 \cdot 10^{x^2}$
   (g) $p(x) = e^{-x^2/2}$
   (h) $q(x) = (2^x + 5)^3$
   (i) $r(x) = \frac{3e^x+2}{e^x-1}$
   (j) $s(x) = (2x - 1)e^{3x}$

# Implicit Differentiation

1. Consider the circle defined by the equation $x^2 + y^2 = 25$:



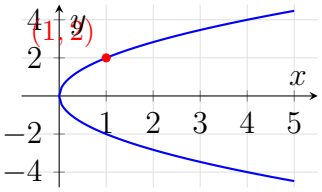

   (a) Find $\frac{dy}{dx}$ using implicit differentiation.

   (b) Find the equation of the tangent line to the circle at the point $(3, 4)$.

   (c) Find all points $(x, y)$ on the circle where the tangent line is horizontal.

   (d) Find all points $(x, y)$ on the circle where the tangent line is vertical.

2. Consider the parabola opening to the right defined by the equation $y^2 = 4x$:



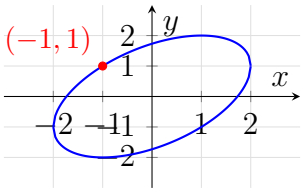

   (a) Find $\frac{dy}{dx}$ using implicit differentiation.

   (b) Find the equation of the tangent line to the parabola at the point $(1, 2)$.

   (c) Explain why this parabola has no horizontal tangent lines.

   (d) Find the point $(x, y)$ on the parabola where the tangent line is vertical.

3. Consider the rotated ellipse defined by the equation $x^2 - xy + y^2 = 3$:



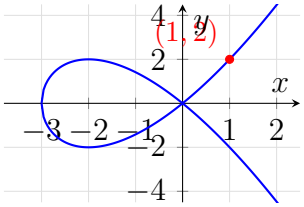

   (a) Verify that the point $(-1, 1)$ lies on the curve, and compute $\frac{dy}{dx}$ in terms of $x$ and $y$.

   (b) Find the equation of the tangent line to the curve at the point $(-1, 1)$.

   (c) Find all points $(x, y)$ on the curve where the tangent line is horizontal.

   (d) Find all points $(x, y)$ on the curve where the tangent line is vertical.

4. Consider the nodal cubic curve (also known as the Tschirnhausen cubic) defined by the equation $y^2 = x^3 + 3x^2$:




   (a) Find $\frac{dy}{dx}$ using implicit differentiation.

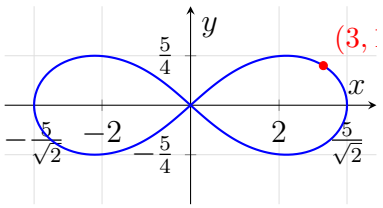
   (b) Find the equation of the tangent line to the curve at the point $(1, 2)$.

   (c) Find all points $(x, y)$ on the curve where the tangent line is horizontal. (Note: at $(0, 0)$, $\frac{dy}{dx}$ is indeterminate $\frac{0}{0}$ because the curve crosses itself.)

   (d) Find all points $(x, y)$ on the curve where the tangent line is vertical.

5. **[Challenge problem]** Consider the lemniscate of Bernoulli defined by the equation:

$$2(x^2 + y^2)^2 = 25(x^2 - y^2)$$




(a) Verify that the point $(3, 1)$ lies on the curve, and use implicit differentiation to show that:

$$\frac{dy}{dx} = \frac{x(25 - 4(x^2 + y^2))}{y(25 + 4(x^2 + y^2))}.$$

(b) Find the equation of the tangent line to the lemniscate at the point $(3, 1)$.

(c) Find the coordinates $(x, y)$ of all points on the lemniscate where the tangent line is horizontal.

(d) Find the coordinates $(x, y)$ of all points on the lemniscate where the tangent line is vertical.